

|          |                                                                                                                                                     |           |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|          |                                                                                                                                                     |           |
| 9(a)(i)  | material introduced into the body<br><b>and</b><br>(position in body) can be detected <b>or</b> absorbed by the tissue (being studied)              | <b>B1</b> |
| 9(a)(ii) | $X = \beta^+ \text{ or } e^+ \text{ and } P = 1$                                                                                                    | <b>B1</b> |
|          | $Q = 0 \quad \text{and } R = 18$                                                                                                                    | <b>B1</b> |
| 9(b)(i)  | positrons (emitted in the decay) and electrons annihilate                                                                                           | <b>B1</b> |
|          | mass of particles becomes energy of gamma photons                                                                                                   | <b>B1</b> |
| 9(b)(ii) | arrival times of photons are processed                                                                                                              | <b>B1</b> |
|          | image built up of tracer concentration in the tissue                                                                                                | <b>B1</b> |
| 9(c)(i)  | $A = \lambda N \text{ and } \lambda = \ln 2 / T$                                                                                                    | <b>C1</b> |
|          | $N = n \times N_A$                                                                                                                                  | <b>C1</b> |
|          | 2 photons produced from each decay, so $R_0 = 2 \times \lambda \times n \times N_A$                                                                 | <b>A1</b> |
|          | $R_0 = (2 \ln 2) n N_A / T$ (allow 0.693 for $\ln 2$ )                                                                                              |           |
| 9(c)(ii) | sketch: exponential decay curve from $t = 0$ to $t = 2T$ , starting at $(0, R_0)$ and with a negative gradient of continuously decreasing magnitude | <b>B1</b> |
|          | line with negative gradient passing through $(T, R_0/2)$ and $(2T, R_0/4)$                                                                          | <b>B1</b> |